Working Together Without Earth: Job Sharing, Team Lifts and Learned Bidding Among Lunar Robots of Different Owners

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Roatify Research · Preprint, October 2026 · Code, data and pre-registration: github.com/azizjonrahimov/Lunar-Haul · Live site: azizjonrahimov.github.io/Lunar-Haul

Abstract. The rovers, cargo vehicles and small robots planned for the lunar south pole belong to different companies, each with its operators on Earth. Sharing vehicles among them shortens waits, but a shared dispatcher on Earth acts only while Earth is above the base's horizon, with gaps of about two weeks. We ask whether the gain survives when the decisions move onto the robots, which hear one another only where terrain allows and lose messages. We simulated a base on the Connecting Ridge with six vehicle classes of three companies, a radio mesh from 5 m terrain and the base's Earth-link windows, under thirteen decision regimes, in 5,480 pre-registered simulated years. At 30 % load, a pooled dispatcher on Earth that stops when Earth sets kept urgent cargo waiting 89.9 h longer than one that never stops; planning ahead before each blackout left 7.3 h of that. An auction run by the robots, at 0.1 message loss, came within 0.8 h of the full-information marketplace, a difference not distinguishable from zero but too wide to show equivalence. A learned linear correction to the bids, with 13 weights, cut the wait by 16.5 % relative to hand-built bids; a 1,537-parameter network trained the same way did not. Pairs of different owners' robots lifting heavy crates raised the share of team items delivered from 12 % to 93 %; a commit step before each meeting cut failed meetings under loss, at a cost in urgent wait. Almost all of the value of sharing, 99.5 % of the gap, is cargo that owners alone never deliver; lifts by two owners' rovers account for 32 % of it. An agreement to share vehicles comes first; dispatch by the robots on the Moon then beats a relay to Earth, because it survives every blackout.

Keywords: lunar south pole, multi-robot task allocation, multi-owner fleets, cooperative transport, intermittent communication, learned bidding

1. Introduction

The first vehicles of a lunar south-pole base will not share an owner. NASA's Lunar Terrain Vehicle task orders of May 2026 went to two companies, Astrolab and Lunar Outpost, for two different rovers [37]. Cargo rovers able to move a tonne or more exist so far only as company designs, small rovers are sold by a third group of firms, and propulsive hoppers by a fourth. Each company plans, drives and supervises its own vehicles from its own control room on Earth. A crate that one company's rover could carry may wait for the owner's own rover to come free, and a crate too heavy for any single rover needs two rovers that may belong to different companies.

Earth is a poor place to coordinate them from. Seen from the south pole, Earth stays within a few degrees of the horizon and terrain hides it for about half of every month [42]. In our previous paper [42] a rover antenna at our base on the Connecting Ridge saw some part of Earth's disc 59 % of the time, in windows of about 16 days separated by gaps of about 11 days, and the oxygen plant 1.2 km away never saw it. NASA's relay programme puts direct-to-Earth coverage at candidate south-pole sites at 17–43 % of a year and coverage with one relay at 80–86 % [38]. A dispatcher on Earth, however good, is blind and mute for days at a time unless someone pays for the relay, and the relay does not reach the oxygen plant either.

Pooling vehicles across owners pays. In our earlier simulation of this base [40], one marketplace that offered every job to every vehicle kept urgent cargo waiting far less than owners working alone. Ride-hailing gives the same result with real data: splitting demand among operators that do not coordinate raises the fleet needed, and the smallest operator pays most [28]. That marketplace, however, assumed a dispatcher that sees every vehicle and every crate at every moment. On the Moon the information sits on the robots, and the robots can talk only to the robots they can see over the terrain. Our question in this paper is whether the gain from sharing survives when decisions move off Earth and onto the robots, under radio links limited by terrain and with messages that get lost.

Each part of this question has been studied alone. Auctions among robots are the standard way to share tasks without a central computer, and their performance under message loss has been measured: the best decentralised allocators lose about 11–13 % of makespan between perfect delivery and 40 % message loss, and settings tuned for perfect links can lose up to 65 % under loss [10, 11]. Those studies use grids or open plains with random or bursty loss, robots of one owner, and compare each algorithm with itself under perfect links. NASA's CADRE rovers will coordinate on the Moon through a leader elected among them [31, 32]. Learned bids inside an auction were published in 2026 [18]. We did not find a study that compares a dispatcher on Earth, limited to the base's real Earth-link windows, with allocation by the robots on the Moon, for vehicles of several owners, over a radio mesh computed from real terrain, and measures the difference in cargo delay (Section 2.7).

We built that comparison on the simulation of [40, 41, 42]. The fleet is six vehicle classes from 2028–2032 programmes, from 80 kg crew rovers to a 2,000 kg cargo rover, split among three owners. Which robot hears which comes from line of sight over NASA's 5 m terrain of the ridge, with a 10 m mast at the base. Earth's visibility from the base comes from the windows of [42], extended to 2046. The decision regimes span the range from owners working alone with conflicts settled by hand on Earth, through one pooled dispatcher on Earth with and without a relay, to three ways of deciding on the Moon: a computer at the base elected as leader, an auction among the robots with hand-built bids, and the same auction with learned bids. The pooled marketplace of [40], given full information and no dependence on Earth, is the yardstick. Crates too heavy for one rover can be lifted by two, from different owners, if both agree over the radio. We wrote the hypotheses, the comparisons and the statistics down before the main runs [43] and changed them five times, three times before the runs they affect and twice after the runs, to correct defects found in them (Section 4.6).

The answer depends on what is shared. At 30 % load, almost the whole gap between owners working alone and the pooled fleet (99.5 % of it) is cargo that owners alone never deliver. Some owners' vehicles cannot lift their own cargo, and only pooling lets another owner's rover carry it (68 % of the gap, together with whatever the pooled dispatcher adds) or a pair of different owners' rovers lift it (32 %) (Section 5.2). Among pooled fleets, where the decision is made matters a great deal. A dispatcher on Earth that stops when Earth sets kept urgent cargo waiting 7.0 times as long as one that never stops, and the difference passed the pre-registered test in each of the six start years in which the base saw Earth less than 60 % of the time (Section 5.3). Uploading plans before each blackout recovered most of the loss, and so did an 80 % relay. Deciding on the Moon left none of it that the data could detect: the robots' own auction, at 0.1 message loss, kept urgent cargo waiting 5.6 % less than the full-information marketplace, a difference whose 90 % interval (−17 % to +5 %) includes zero (Section 5.4), and a learned correction to the bids cut the wait by 16.5 % relative to the auction (Section 5.6).

Two pre-registered expectations failed, and we report them as they came out. We expected the robots' auction to come within 10 % of the full-information marketplace; the test could not show it. The difference was −0.84 h with a 90 % interval of −2.6 to 0.7 h, which reaches past the margin of ±1.5 h on at least one side, so the auction

may be better or worse than the marketplace by more than 10 %, and the data cannot tell. We expected a small neural network to carry any gain from learning; it did not beat the hand-built bids at any load, and a linear model with the same inputs carried all of it. We also expected learning to help nothing at light load, and there the data could show neither a gain nor its absence: the gain's interval (−0.1 to 2.2 h) includes zero and reaches past the ±2 % margin.

Our contributions are:

- **A measured cost of keeping the dispatcher on Earth.** For one south-pole base, the urgent-cargo delay of a pooled fleet dispatched from Earth only while Earth is in view, against the same fleet dispatched without interruption, per start year from 2028 to 2046, with the share that planning ahead and a relay recover.
- **A comparison of decision places on a realistic robot network.** Owners alone, a pooled dispatcher on Earth, an elected leader on the Moon and an auction among the robots, on the same seeds, over a radio mesh computed from 5 m terrain, with message loss, bundle depth and radio range varied one at a time.
- **Cross-owner team lifts under message loss.** How much cargo only becomes movable when two owners' robots lift together, and what a meeting protocol with a commit step and a timeout costs when messages are lost.
- **An honest test of learned bidding.** A learned correction to the auction's bids, trained by imitation and by search on the simulated objective, evaluated once on held-out seeds against hand-built bids, a linear model with the same inputs and six conditions outside its training range.

Several things are outside this paper. Owners here report their costs honestly and accept the assignments they win; payments, gain sharing and owners who misreport are the subject of a companion paper [44]. Rules of the road, separation and the safety check that every vehicle runs before it moves are the subject of another [45]; here vehicles do not interfere with one another on the road. The radio model is geometric: a link exists when two antennas see each other within a set range, with no link budget, and the range and clearance are provisional values that we vary. Robots that nobody hears drive to a rally point that the base's mast can see; that rule is our assumption, and without it the robots' network falls apart on this terrain (Section 3.9). The full-information marketplace uses the same bid rule as the hand-built auction with perfect knowledge; it is not an optimal dispatcher, and two of our regimes beat it. One base layout, one terrain tile and one demand model are simulated, at three levels of demand of which the lightest is the closest to today's plans.

2. Related work

2.1 Sharing tasks among robots

Gerkey and Matarić classified multi-robot task allocation by whether a robot does one task or several at a time, whether a task needs one robot or several, and whether tasks are assigned once or over time [1]. Our problem has single-task robots, tasks that need one robot or two, and assignment over time as crates land. For the instantaneous single-robot case the Hungarian method gives the optimum in polynomial time, and greedy assignment is within a factor of two of it. The time-extended version is strongly NP-hard. Gerkey and Matarić also judged that for fewer than about 200 robots a centralised assignment over a broadcast network is “likely the better choice” [1], which is the case for one base with a dozen robots, provided a central computer can hear them all.

Auctions are the standard way to share tasks without one. In a sequential single-item auction every robot bids its cost of adding each open task to its plan, the lowest bid wins one task, and the round repeats. Its total cost is at most twice the optimum when travel costs obey the triangle inequality, and in experiments it comes close to combinatorial auctions [2, 3]. The consensus-based bundle algorithm (CBBA) lets each robot build a bundle of several tasks greedily and then settle conflicts with its neighbours only; it is conflict-free and tolerates inconsistent information, and it guarantees half the optimal score when each added task is worth no more than the one before [4, 5]. An asynchronous version drops the global rounds [6]. Market-based architectures such as TraderBots let robots trade tasks to raise their own profit, with local leaders planning centrally when that pays [7]. Our on-Moon auction is a sequential single-item auction whose winning bids are rebroadcast, with CBBA's rule for settling conflicts and an optional bundle depth (Section 3.7).

2.2 Allocation when messages are lost

Otte, Kuhlman and Sofge compared six auctions under random and bursty message loss, in simulation and on quadrotors [8]. The ranking of the auctions changed with the loss rate, single-round auctions did worst because a robot that misses the only round sits idle, and rebroadcasting earlier winners in later rounds helped every multi-round auction. Nayak and colleagues compared five decentralised allocators under three loss models and found that all degrade, and that the best algorithm at high communication is not the best at none [9]. Lott and Honary measured the bundle depth of CBBA-type allocators under 25 % loss: the asynchronous CBBA's penalty in total travel grew from 19.7 % at depth 1 to 64.7 % at depth 12, and the best depth moved from 12 with perfect links to 2 under loss [10]. Their second benchmark of six allocators over 25 communication conditions found the best methods losing about 11–13 % of makespan between perfect delivery and 40 % loss [11]. Algorithms for very low communication exist [12].

These studies share a design. Robots move on a grid or an open plane, links fail at random or in bursts, every robot belongs to one team, and each algorithm is compared with itself under perfect links, so the cost of decentralisation relative to a central dispatcher is not measured. Lott and Honary also showed that settings chosen under one communication condition can be wrong under another, and that tuning and evaluation need separate scenarios [10]; we follow that rule (Section 4.3).

2.3 Tasks that need two robots

Assigning tasks that need several robots at once is a set-partitioning problem, strongly NP-hard and in general hard to approximate [1, 14, 15]. Korsah, Stentz and Dias add the dependencies between schedules that make it harder in practice: when two robots must meet, the task's value depends on both robots' plans [13]. Market methods handle complex tasks by letting robots bid on decompositions [16], and a coupled-constraint extension of CBBA handles tasks whose value depends on when or whether another task is done [5]. Two-rover cargo carrying has been demonstrated on lunar-analogue terrain: two 800 kg rovers carried a 475 kg, 5.5 m load with a mean separation error of 9.2 cm, over 13.8 KB/s of traffic between them [17]. Those rovers belonged to one team and coordinated at the level of control. With a dozen robots and lifts by pairs there are few pairs to enumerate. The hard part is the meeting: both robots must arrive, and under message loss one can wait for a partner that never comes. We found no study of lifts by robots of different owners, or of what a meeting protocol costs under message loss at the level of a logistics operation.

2.4 Learned allocation

Learned allocators either output assignments directly, such as graph-attention policies trained by reinforcement learning or imitation [21, 22], or learn scores that a classical solver then assigns. In the second group, a learned incentive feeds the Hungarian method [20], a graph network approximates CBBA's score [19], and a 2026 study replaced CBBA's hand-built bid with a policy trained by proximal policy optimisation while keeping the auction and consensus phases [18]. That study is the closest prior work to our learned bids; its robots are cooperative and its links perfect. Ride-hailing dispatch at scale adds a learned value of a driver's position to a combinatorial matching, and its deployed gains are in the low single digits of percent: over 1.3 % more driver income in A/B tests and up to 5.3 % in major metrics after deployment [23]. Neural solvers for routing problems generalise poorly beyond the sizes they were trained on [25], and operations researchers have asked for evaluations against strong classical heuristics on the same instances and hardware [26]. Imitation of an expert by dataset aggregation (DAgger) avoids the compounding errors of plain cloning [24]. We use DAgger first and then search on the simulated objective, keep the auction unchanged, and test against a linear model with the same inputs (Section 3.10).

2.5 Fleets of several owners

Robotics allocation usually assumes cooperative robots of one owner. Transport research has measured what fragmentation among owners costs. Splitting the demand for mobility on demand among platforms that do not coordinate raises the cost of rebalancing, and unequal market shares make it grow faster [27]. With data from five cities, Kondor and colleagues found that each extra operator raises the fleet needed by a factor that ranges from 2.5 % of demand in Manhattan to 67 % in Curitiba, and that in Singapore with shares of 50, 35 and 15 % the smallest operator's fleet grows by 48 % against 8.5 % for the largest [28]. In freight, joint route planning among carriers saves about 20–30 % [51], and carriers rank a trusted coordinator and a fair split of the benefits as the main obstacles [29]. Open-RMF lets robots of unrelated vendors share a traffic schedule and bid for tasks across fleets without changing their own fleet managers [30]; it is the closest working system to what we simulate, and it assumes a building network that is always up.

2.6 Lunar surface autonomy and networks

CADRE, NASA's three-rover technology demonstration, runs a central planner on a leader elected among the rovers and the lander's base station, with execution on every rover; the leader is re-elected at every 30-minute wake cycle [31, 32]. One owner and three rovers explore; there is no cargo. A 2024 lunar variant of the consensus-based auction models payload, priority, lunar night and cooperative tasks for one owner's robots [33], and an auction hierarchy has been simulated for regolith mining [34]. In the DARPA Subterranean Challenge, robots relayed data for one another through mesh radios, dropped relays and carried data out of range [35]; the winning team deployed single-robot autonomy only, citing the field testing that multi-robot coordination would need [36]. Lunar surface traffic has been raised as a regulatory question without algorithms [50].

NASA's LunaNet specification standardises interoperability among lunar users at the network layer, with Internet protocols and the Bundle Protocol version 7 for delay-tolerant networking; the radio link between two surface vehicles is left to bilateral agreement [39]. Delay-tolerant networking became an operational NASA service in 2026 [49]. The relay programme states its case as coverage: 17–43 % a year at candidate south-pole sites from direct-to-Earth links alone, 80–86 % with one relay [38]. None of these documents says how vehicles of different owners should share work, or what happens to a shared operation while the base cannot see Earth.

2.7 Our earlier work and the gap

Our first paper built a discrete-event simulation of cargo movement at a south-pole base and showed that one marketplace offering every job to every vehicle keeps cargo waiting far less than owners who use only their own vehicles [40]. The second added landing precision, pad standoff, a plume keep-out and rover autonomy on NASA's 5 m terrain of the Connecting Ridge [41]. The third computed the base's Earth-link windows from the ephemerides of SPICE, NASA's planetary geometry toolkit, and laser-altimeter terrain and priced their gaps for a fleet driven or supervised from Earth [42]. All three assumed one dispatcher that sees everything.

Table 1 sets the closest works against the features our question needs. We did not find a study that compares allocation by robots on the Moon with a shared dispatcher on Earth limited to real terrain-masked Earth-link windows, for vehicles of several owners, over a radio mesh computed from real terrain, with lifts by pairs of robots from different owners, measured in cargo delay. Each feature on its own is published; the comparison is what this paper adds.

Table 1. Features of the closest prior work. "Yes" means the feature is part of the study's model or experiment; "–" means it is absent. Terrain radio means links computed from line of sight over a terrain model.

Work	Several owners	Message loss	Terrain radio	Earth-link windows	Two-robot tasks	Learned bids	Compared with a central dispatcher	Logistics outcome
Otte et al. 2020 [8]	–	Yes	–	–	–	–	–	–
Lott and Honary 2026 [10, 11]	–	Yes	–	–	–	–	–	–
Whitten et al. 2011 [5]	–	–	–	–	Yes	–	–	–
Krawciw et al. 2026 [17]	–	–	–	–	Yes (hardware)	–	–	–
Rodriguez et al. 2026 [18]	–	–	–	–	–	Yes	–	–
Kondor et al. 2022 [28]	Yes	–	–	–	–	–	Yes	Yes (fleet size)
Open-RMF [30]	Yes	–	–	–	–	–	–	–
CADRE [31, 32]	–	–	–	–	–	–	–	–
Lunar consensus auction, IAC 2024 [33]	–	–	–	–	Yes	–	–	–
Our earlier work [40, 42]	Yes	–	–	Yes [42]	Yes [40]	–	Yes	Yes
This paper	Yes	Yes	Yes	Yes	Yes, across owners	Yes	Yes	Yes

3. Model

The model extends the discrete-event simulation of [40, 41, 42]: time jumps from one event, such as a landing, the end of a drive or a lost message being sent again, to the next. Everything that was in those papers and is not described here is unchanged, and with the new features switched off the engine reproduces their results exactly (Section 3.11). Every input carries a flag at first use, as in [41, 42]: **Measured** (flight, laboratory or survey data), **Spec** (a published design value or company claim), **Modelled** (produced by a model, including ours) or **Assumed** (our choice, varied where it matters). Results produced by this study are never flagged; they come from the result files listed under Availability.

3.1 Base, terrain and demand

The base is the one of [41, 42], on the Connecting Ridge between Shackleton and de Gerlache craters, on NASA Goddard's 5 m terrain tile from the Lunar Orbiter Laser Altimeter (Measured). Its sites are the habitat and control centre (BASE), a cargo depot, a power plant, a construction site, an oxygen pilot plant, a science site and a

permanently shadowed crater; landers use two pads 1 km from the base. Driving times, energy and routes come from the driving tables of [41].

Demand follows the landings and bulk flows of [40], scaled by a factor k so that the two cargo-class vehicles are busy a set share of the time: about 10, 30 and 60 % (k = 0.857, 2.806 and 6.220, Assumed; calibrated on five seeds to within one percentage point of the target). We call these the three loads. The two cargo-class vehicles are the only ones that can lift most of the cargo, so the share of the whole fleet's time spent working is much lower: a share reported in Section 5.1. Landed cargo belongs to the owner who ordered it, a third each. Bulk regolith moves in 500 kg lots (Assumed), split among owners in the same shares. A landed item becomes a team item with probability 0.10 (Assumed; 0, 0.10 and 0.25 in Section 4.2), drawn from its own random stream, and then weighs between the largest single vehicle's capacity and the largest pair's, 2,000–2,800 kg. Items heavier than any pair can lift go on a separate list, “needs heavy-lift equipment”, and are never offered to the robots; they are reported and excluded from every delay. Each run simulates one year from an empty base.

3.2 Vehicles and owners

Table 2 lists the six vehicle classes in the simulated fleets. They are the two rovers NASA ordered in May 2026 for its Lunar Terrain Vehicle service [37], two cargo-class rovers offered by the same companies, a small rover and a propulsive hopper; a seventh class, a smaller hopper, is in the vehicle table but in none of the fleets simulated here. The crew rovers carry little cargo: 80 kg each, and 10 km on a charge. The cargo rovers carry 2,000 and 800 kg. The small rover carries 200 kg and the hopper 10 kg. Three owners share eight vehicles: owner A has the crew rover CLV-1 and the cargo rover FLEX, owner B the crew rover Pegasus and the cargo rover Eagle, and owner C two small rovers and two hoppers. Owner C therefore cannot carry most of its own cargo, and of the pairs a single owner can field, only A's FLEX with CLV-1 (2,080 kg) can lift any team item. Hoppers fly straight hops at 20 km/h on a 50 km propellant budget over 14 days and then retire (Spec for the budget and window, Assumed for speed); a variant refuels them every month. The full table, with every source, is in the repository.

Table 2. Vehicle classes in the simulated fleets. Speed is the modelled average, 0.6 of the published top speed where only that is given (Assumed, as in [41]). Energy per kilometre comes from the traction model of [41] at the vehicle's mass (Modelled). Battery capacities are Assumed except where a range per charge is published.

Class	Owner	Mass (kg)	Payload (kg)	Speed (km/h)	Energy (Wh/km)	Battery	Works at night	Can join a team lift
Astrolab CLV-1 (crew rover)	A	907, Spec	80, Spec	5.8	75.8	758 Wh (10 km per charge, Spec); 6 km a day, Spec	No, Assumed	Yes, Assumed
Astrolab FLEX (cargo)	A	1,600, Assumed	2,000, Spec (company claim)	6, Assumed	133.7	40 kWh, Assumed	No	Yes
Lunar Outpost Pegasus (crew rover)	B	700, Assumed	80, Spec	8.7	58.5	585 Wh (10 km per charge); 6 km a day, Spec	No	Yes
Lunar Outpost Eagle (cargo)	B	1,500, Assumed	800, Assumed	6, Assumed	125.4	40 kWh, Assumed	Yes, Spec	Yes
Small rover (MAPP class), two	C	250, Spec	200, Spec	6.5	20.9	2 kWh, Assumed	No, Assumed	No
Hopper (MoonFall class), two	C	250 wet, Spec	10, Assumed	20, Assumed	propellant: 50 km in 14 days, Spec	–	No	No

3.3 The link to Earth

Earth is in view of the base when any part of Earth's disc is above the terrain horizon of a 2.5 m antenna at BASE, the criterion of the published visibility maps [42]. The windows for 2028–2031 are those of [42]; for start years from 2032 we use the same pipeline's windows extended to 2046 (Modelled). A relay, where a regime has

one, adds link time as an on–off process with a stated coverage, independent of the direct link, with the gap pattern of [42] (Assumed). Earth's visibility from the base differs by year: in the start years we simulate it ranges from 54.7 % to 62.6 % of the year, and in six of them it is below 60 %, the threshold we fixed for the availability test (Section 4.5).

3.4 The robots' radio network

Two robots can talk directly when their antennas see each other over the terrain and are within range. We computed line of sight for every pair of 2,421 points (the sites, the pads and samples along every route) on the 5 m terrain with the Moon's curvature, the same arithmetic as [41]. Antennas are 2.5 m high on rovers and 1.0 m on small robots and hoppers (Assumed), and a 10 m mast at BASE carries the base computer (Assumed, after NASA Glenn's surface relay concept as in [42]). The range is 5 km and the clearance margin 0 m (both Assumed; 2, 10 km and unlimited range in Section 4.2). A message can travel over several hops through robots or the mast.

Line of sight is rare on this ridge. Two rover antennas 2–5 km apart see each other in 9.31 % of point pairs, and the mast sees rovers directly only within 2 km of the base. Rovers nevertheless form one connected network from 2 km range upward, and every point reaches the mast in at most 4 hops at 5 km range; small robots leave four points cut off. The robots' positions are interpolated along their committed drives, and the network is evaluated whenever a decision is made.

3.5 Messages

Every hop of a message is lost independently with probability p (Bernoulli loss, Assumed) or in bursts (a two-state Gilbert–Elliott channel with mean burst length 10 messages and the same long-run loss, Assumed), and takes 10 ms (Assumed). The on-Moon regimes run at $p = 0.10$ unless stated; Section 4.2 varies it from 0 to 0.40. A robot whose message was lost is reconsidered 3 minutes later (Assumed), with at most one such retry pending at a time. We count transmissions, messages delivered and lost, and messages per robot-hour.

3.6 Decision regimes

A regime says who decides which vehicle takes which job, where that decision is made, and when it can be made. Table 3 lists them. The owners-alone regimes use the in-house dispatcher of [40]: each owner assigns only its own cargo to its own vehicles, and its own pairs may lift together. When two owners' vehicles want the same apron at the same time, the newcomer waits 2 h of hand coordination (Assumed), and only while Earth is in view. The pooled regimes use the marketplace of [40], which offers every job to every vehicle in auction rounds and forms team lifts from any pair; they differ in where it runs.

A dispatcher on Earth acts only while Earth is in view of the base. Under **strict** fallback, vehicles finish their current job and wait when Earth sets. Under **plan ahead**, the dispatcher uploads at loss of signal a queue per vehicle for the items already waiting, and a standing rule for cargo that will land during the blackout: oldest first, nearest free vehicle, one item per trip. Items created on the Moon during a blackout, such as returns, wait for Earth.

On the Moon, the **elected leader** regime runs the same marketplace on the base computer at the mast, over the robots in its part of the network. The base computer fails at random (mean time between failures 2,000 h, repair 24 h, both Assumed); then, after a 30-minute election (Assumed, after CADRE's wake cycle [31]), the robot that reaches most others leads until the base computer returns. Status and assignment messages travel over the lossy network. The **auction** and **learned** regimes are described next.

Table 3. Decision regimes. “When” is when assignments can be made. The full-information marketplace is the reference: the same bid rule as the pooled regimes, with perfect knowledge and no dependence on Earth or on the radio network. It is not an optimal dispatcher.

Regime	Code (Tables 5–8)	Jobs shared across owners	Who decides, and where	When	Team lifts
Owners alone, from Earth, strict	A_EARTH strict	No	Each owner's control room on Earth	Earth in view	Own pairs
Owners alone, from Earth, plan ahead	A_EARTH plan_ahead	No	Each owner, with uploads before blackouts	Always (plans age)	Own pairs
Owners alone, from Earth, full coverage	A_EARTH full coverage	No	Each owner, with Earth always in view	Always	Own pairs
Owners alone, on the Moon	A_MOON	No	Each owner's robots, within its part of the network; apron conflicts still go to Earth	Always	Own pairs
Pooled, from Earth, strict	B_EARTH strict	Yes	One dispatcher on Earth	Earth in view	Any pair
Pooled, from Earth, plan ahead	B_EARTH plan_ahead	Yes	One dispatcher, with uploads before blackouts	Always (plans age)	Any pair
Pooled, from Earth, relay 0.8 / 1.0	B_EARTH_RELAY 0.8 / B_EARTH_RELAY 1.0	Yes	One dispatcher, through a relay of coverage 0.8 or 1.0	Earth or relay in view	Any pair
Elected leader on the Moon	C_LEADER	Yes	Base computer, or a robot when it is down	Always, within the leader's part of the network	Any pair
Auction on the Moon	C_AUCTION	Yes	Every robot, by bidding	Always, within each part of the network	Any pair, with commit and timeout
Learned bids (network or linear)	C_LEARNED_MLP / C_LEARNED_LIN	Yes	As the auction, with a learned bid correction	As the auction	As the auction
Full-information marketplace	O_ORACLE	Yes	One dispatcher that sees everything	Always	Any pair

3.7 The auction among the robots

Each connected part of the network runs the marketplace's own rounds. A part that contains the mast knows every waiting item; any other part knows only the items at sites where one of its robots stands. In a round, each robot computes its best bid, the marginal cost in hours of adding the job to its plan, weighted by the item's urgency as in [40], and floods it to the robots it can reach, losing it hop by hop with the channel's probability. The true best bid always wins its item. A robot that did not hear the best bid and believes its own is best wins in parallel if its items do not overlap with the winner's; if they do, CBBA's rule settles the conflict when the winner's rebroadcast reaches it within the remaining rounds [4], and otherwise the robot drives to the pad for nothing. The number of rounds is at most the larger of the number of items and of robots, times the network's diameter in hops, so that a perfect network never cuts the auction short.

With bundle depth greater than one, a CBBA-like variant, robots also bid for up to depth − 1 later jobs from the end of their commitments. A claim starts when the robot comes free if its items are still waiting, and is dropped otherwise. Claims are hidden from robots in the same part of the network. With perfect links and depth 1, the auction reproduces the marketplace's event trace exactly (Section 3.11).

3.8 Team lifts across owners

A team item is lifted by two vehicles together. In the pooled regimes, the auction pairs any two vehicles that can join a team lift and together can carry the item; under the owners-alone regimes the pair must have one owner. The pair agrees by a two-phase commit over the radio: the proposer sends a proposal, the partner answers yes,

and the proposer sends a commit. A lost proposal or answer means no team in this round. A lost commit sends the proposer alone to the pad, where it waits the timeout, 1 h (Assumed; 0.5 and 2 h in Section 4.2), and gives up; the item returns to the pool and the failed meeting is counted with its waiting time. Coupling at the pad takes 1 h, as in [40]. Some message is always the last one, so no protocol over a lossy channel can guarantee that both robots know the other will come (the Two Generals problem); the commit step trades failures of one kind for another, and Section 5.7 measures the balance. An exploratory variant lets the base computer arbitrate meetings within its part of the network.

3.9 Robots that nobody hears

Line of sight on the ridge is so sparse that a robot finishing a delivery at the oxygen plant, the science site, the shadowed crater or pad B can end up where no other robot and no mast hears it. Without a rule it is never reached again. Our lost-link rule (Assumed) sends an idle robot that nobody hears to the nearest site that sees the mast at its antenna height (to BASE for owners alone on the Moon), where it waits to be heard. Section 5.5 reports what switching the rule off costs in the E41 experiment (Section 4.2). We added the rule during development, after a five-seed run in which the elected leader without it delivered 58 t of cargo in a year against 1,487 t with it (Section 4.6). Section 4.2 includes the rule switched off, and a second 10 m mast at pad B, as network-design variants.

3.10 Learned bids

The learned regimes keep the auction unchanged and add a correction to each bid: the robot bids $b' = b - u \cdot g(x)$, where b is the hand-built bid, u converts hours of urgent wait into the bid's units as in [40], and g is a learned function of thirteen features that a robot knows locally. They are the planned hours of the job, the item's age, priority and slack to its target time (24 h for urgent items, as pre-registered), the load as a share of capacity, the hours to drive to the pad and the battery's state of charge. They also include the hours until the robot's next dark period or until its battery would run out (whichever comes first), whether it is in the mast's part of the network, how many robots it hears, how many claims it holds, the number of items in the bundle, and the hours until the next published landing at the item's pad or the robot's site. The last two time features let a robot anticipate what the hand-built bid ignores. The network never assigns anything, so it cannot create a conflict or an infeasible plan that the auction would not.

We trained two forms of g: a multilayer perceptron with two hidden layers of 32 units (1,537 parameters), and a linear model with 13 weights. Training used seeds 1000 and up, selection used tuning seeds 2000 and up, and the evaluation seeds 1–30 were used once (Section 4.3). The first stage imitated the full-information marketplace by dataset aggregation [24]: each of 37,941 training rounds (624,640 robot-item rows) was labelled with the item the marketplace would give each robot. Section 5.9 reports what that stage produced and how often the hand-built bids already chose the marketplace's item. The marketplace uses the same bid rule with full information, so imitating it reproduces the hand-built bid.

The second stage searched for g on the objective itself with the cross-entropy method: candidates were scored by mean urgent wait at the middle load and message loss 0.10 on four training seeds, with common random numbers across candidates, and each generation's best was scored on three tuning seeds; the generation with the best tuning score was kept. The linear model searched its 13 weights over 6 generations of 10 candidates; the network searched only its 32 output weights over 5 generations of 12, with its hidden layers left at their initial values because imitation never moved them. Every candidate's score is recorded in the policy files. We decided before the main runs to carry both models into every experiment (Section 4.6).

3.11 Checks

Each regime was checked against cases whose answer is known. The pooled dispatcher on Earth with coverage 1.0 reproduces the full-information marketplace's event trace and summary exactly; so does the elected leader with perfect radio and no base failures; so does the auction with perfect links and depth 1; and owners alone with full coverage, no hand-coordination time and no own-pair lifts reproduce the in-house dispatcher of [40]. With every new feature off, 20 result rows of [41, 42] rerun with no difference in any numeric column, and the test suite (547 tests) passes. On 200 small random instances the sequential auction's total cost was at most 1.255 times the optimum from the Hungarian method (worst-case bound 2 [2]), and the CBBA-like variant's score at least 85.1 % of the brute-force optimum (bound 50 % [4]); no item was ever assigned twice. These instances satisfy the bounds' assumptions, so the result checks the implementation and says nothing new about the algorithms. In addition, 107 checks of the engine's phase-5 verification suite (`lunarsim/vv5.py`, results in `results/vv_phase5.json` and on the live site's verification page) bear on this paper: 89 verifications that the code does what the model says, eighteen validations against references outside the model, by Sargent's technique analytic (24), degenerate tests (16), historical data (16), internal validity (15), comparison to other models (14), traces (12), metamorphic testing (5), extreme condition (4), event validity (1). Deterministic identities are held to 10^{-9} relative or exact; stochastic checks pass when the error is within a tolerance set in advance, and a 99 % batch-means interval that misses the reference is recorded as a bias flag, not a failure; the checks ran at commit `5c5b8fd`. Their outcomes are in Section 5.9.

4. Experiment design

4.1 Questions and hypotheses

We pre-registered the hypotheses [48] in a dated file in the repository before the main runs [43]. The confirmatory family, tested at the middle load with Holm's correction, is:

- **H1, sharing.** Pooled regimes with full Earth coverage cut the mean urgent wait relative to owners alone with full coverage, at 30 and 60 % load.
- **H2, availability.** A pooled dispatcher on Earth with strict fallback keeps urgent cargo waiting longer than the full-information marketplace, by more than 10 % of the marketplace's mean (the lower end of the 95 % interval must clear that margin), in each start year whose base visibility is below 60 % and pooled over years. Secondary: planning ahead recovers part; relay 0.8 comes within 10 % of the marketplace by an equivalence test.
- **H3a, decentralisation.** At message loss up to 0.10, the robots' auction is within ±10 % of the full-information marketplace (two one-sided tests [46, 47]). Exploratory: deeper bundles do worse at loss 0.25–0.40 (H3b); the elected leader matches the auction unless the network splits (H3c).
- **H4, learning.** At 30–60 % load, learned bids beat hand-built bids on the held-out seeds; at 10 % load the difference is within ±2 %. The linear model and every test outside the training range are reported whichever way they come out.
- **H6, team lifts.** Allowing cross-owner pairs raises the share of items delivered; at loss 0.25 and above, failed meetings rise without the commit-and-timeout step.

H5 (payments) and H7–H10 (safety, Mars, sizing) belong to the companion papers [44, 45].

4.2 Experiments

Table 4 lists the three experiments. All use three equal owners, team share 0.10 and message loss 0.10 for the on-Moon regimes unless the experiment varies them. E40 crosses the regimes with the three loads. E41 varies one factor at a time around the auction at the middle load: message loss, bundle depth under loss, radio range, the lost-link rule and a second mast. E42 varies the team-lift share, whether pairs may cross owners, the loss rate and the meeting protocol; it also runs the owners-alone regime at each team share, as reference rows (Table 8) that the split below does not use. H1 is split into two parts, both paired by seed on the first block. The dispatch part compares the auction with pairs restricted to one owner against E40's owners-alone cell with full Earth coverage (single-robot jobs pooled across owners, together with the pooled dispatcher). The team part compares the auction with cross-owner pairs against the auction with same-owner pairs (lifts by pairs of different owners' rovers).

Table 4. Experiments. Runs are simulated years. Seeds are crossed with start years as described in Section 4.3.

Experiment	Factors and levels	Seeds per cell	Cells	Runs
E40, main comparison	12 regimes × load 10, 30, 60 %; owners alone with full coverage at 30 and 60 %; hopper refuelling for the full-information marketplace and the auction at 30 %	80 (30 + 50); refuelling 30	40	2,750
E41, network and messages	Around the auction, elected leader and both learned regimes at 30 %: loss 0, 0.10, 0.25, 0.40 and bursty 0.25; depth 1, 2, 5 × loss 0.25, 0.40 (not the leader, which does not bundle); range 2, 5, 10 km and unlimited; lost-link rule off; second mast at pad B	30	52	1,560
E42, team lifts	Auction at 30 %: team share 0, 0.10, 0.25 × cross-owner pairs no/yes × loss 0, 0.10, 0.25, 0.40, with commit and a 1 h timeout; at share 0.25: timeout 0.5 and 2 h, no commit, base-arbitrated meetings (exploratory); owners alone at each share	30	39	1,170

4.3 Seeds, start years and common random numbers

A seed fixes the landings, the manifests, the item masses, the team draws, the base computer's failures and the message losses, so all regimes face the same year when they share a seed. Evaluation uses two blocks. Block one is seeds 1–30 with start years 2028 to 2031 in rotation (start year 2028 plus the seed modulo 4), on the link windows of [42]. Block two is start years 2032, 2035, 2038, 2041 and 2044, each with seeds 1–10, on the extended windows. E40 uses both blocks, 80 runs per cell; E41 and E42 use block one. Training used seeds 1000 and up and tuning seeds 2000 and up; no evaluation seed was used for either.

4.4 Outcomes

The primary outcome is the mean waiting time of urgent cargo in a run, in hours, from the moment the item is ready to its pickup. Items still waiting at the end of the year count with their wait so far (censored at the horizon); cargo on the heavy-lift list is excluded. The secondary outcomes are the 95th percentile of urgent wait, the share of urgent items picked up within 24 h, tonnes delivered and failed team-lift meetings; the message and driving measures are reported in [44]. Because censored means mix waiting with not moving at all, we also report the mean over delivered items only, labelled as such.

4.5 Statistics

Every comparison is paired by seed and start year. The pairing holds for demand, landings and initial faults: in the engine's phase-5 checks (D10, Section 3.11) every regime's E40 run had the same demand as the oracle's on the same seed and start year across 880 regime pairs; the per-job intervention and fault redraws are shared in draw order and diverge once the allocations differ, in every regime. Against the oracle, pairing cut the variance of the paired difference in mean delay per delivery to 0.00–0.52 of the unpaired sum for B_EARTH plan_ahead, B_EARTH_RELAY 0.8, B_EARTH_RELAY 1.0, C_AUCTION, C_LEADER, C_LEARNED_LIN,

C_LEARNED_MLP, and only to 0.90–0.99 for A_EARTH plan_ahead, A_EARTH strict, A_MOON, B_EARTH strict, whose delay varies mostly between start years; their paired intervals are little narrower than unpaired ones would be, and the bootstrap over pairs includes that variation. We report the mean paired difference with a percentile bootstrap 95 % interval from 20,000 resamples of the pairs [53]. The confirmatory family {H1, H2, H4 (linear), H4 (network), H6} at the middle load is corrected with Holm's procedure [52] (Table 7); H3a is an equivalence claim tested separately (Section 5.4); all other intervals are per comparison and unadjusted. A claim of no practical difference needs two one-sided tests at a margin of ±10 % of the full-information marketplace's mean (±2 % for H4 at light load), using the 90 % interval [46, 47]. For H2 the comparison is made in each start year below 60 % visibility and pooled; the block of start years is reported, and the pooled interval includes the variation between years. The rule for adding seeds (blocks of 10 up to 60 if the half-width exceeded 10 % of the mean) was fixed in advance. No seeds were added: every cell holds the 30 seeds of the design (and the 50 runs of the second block where it applies). The rule was not applied to H3a's comparison, whose 90 % half-width of 1.64 h exceeds 10 % of the marketplace mean (1.49 h); by the rule ten more seeds should have been run for it. We report the equivalence test as it stands rather than extend the sample after seeing the result (Section 7).

4.6 Choices made after looking at results

Five changes were made to the registered plan [43]. First, the owner of the study regrouped the planned papers; the hypotheses of this paper were unchanged. Second, after a five-seed development run showed that the gap between owners alone and the full-information marketplace was almost all cargo that owners alone cannot move, we replaced H2's yardstick (“half that gap”) with the comparison inside the pooled fleet given above; the original yardstick is still reported. Third, after the learned models had been evaluated once on seeds 1–30, we decided to run both in every experiment instead of choosing one, and froze both. The lost-link rule (Section 3.9), the hopper refuelling variant, the decision to let owners alone pair their own robots, and the owners-alone regime with full coverage (which H1 names but the first cell list lacked) were set during development, before any main run of the cells they affect. A fourth change was made after the main runs and before the final analysis, when two defects were found in the runs. Regimes that never wait for Earth give the same run for a seed whatever the start year, so their second block (five start years, seeds 1–10) repeated runs; every comparison between two such regimes therefore uses the first block only, as the registered design states, and an interim statement that the auction beat the marketplace, which counted the repeats as independent runs, is withdrawn. The auction had ordered robots by a per-process hash, so its runs under message loss were valid draws but not bit-reproducible; the order was fixed and the affected runs were repeated, with the old rows archived and compared with the new ones. Both are dated in the pre-registration file [43]. The fifth change was made after the first full analysis and the second draft of this paper, when a defect in the engine's event handling was found while timing the largest cells of a later experiment: The dispatcher's consolidation hold re-posted a wake-up for every held lane on every call, so the event queue grew with fleet size and, because one dispatch call does not reach a fixed point, the duplicates also changed assignments. The fix, pre-registered as change 5 before any rerun row, posts one wake-up per lane and lot and repeats dispatch at the same instant until nothing changes. Every block of E40, E41 and E42 was rerun with it; the 4,052 earlier rows are archived and reproducible with the fix off, and every number in this paper is from the rerun rows. Against the archived rows with the same cell and seed (E40: 2,750 rows matched, the primary outcome moved by −0.01 h on average and 1.3 h in absolute value, at most 105 h in one run; E42: 1,170 rows matched, the primary outcome moved by −0.02 h on average and 0.3 h in absolute value, at most 6 h in one run; E41: 132 rows matched, the primary outcome moved by +0.04 h on average and 0.3 h in absolute value, at most 3 h in one run). The per-cell before/after tables are results/E40_fixpoint_before_after.csv and the E41 and E42 files beside it.

4.7 Compute

Runs were made on one laptop with 8 GB of memory, one process at a time per experiment, in chunks of at most 540 s that resume from the last finished run. A simulated year took 0.3 s to 71 s depending on the regime and load, with at most 360 MB of memory. The learned models' training took 1.8 h in all.

5. Results

5.1 What each regime delivers

At every load the regimes fall into three groups (Table 5). Owners working alone keep urgent cargo waiting longest, whether they are coordinated from Earth or on the Moon: 1,217 to 1,257 h on average at the middle load. Pooled regimes that lose contact with Earth part of the time come next, from 17.6 h with an 80 % relay to 104.9 h with strict fallback. The regimes that decide on the Moon, the full-information marketplace and the dispatcher on Earth with a relay that is always up sit together at the bottom, from 11.8 h with learned linear bids to 14.9 h with the relay that is always up and the full-information marketplace. Tonnes delivered are almost the same within the pooled group (1,454 to 1,471 t) and much lower for owners alone (943 to 956 t).

The fleet is lightly used even at the middle load. The two cargo rovers are busy 30 % of the year, the whole fleet 7.6 %, because the crew rovers, small rovers and hoppers can lift almost none of the cargo. At light load every pooled regime keeps urgent waits within 7.9 h of one another except the strict dispatcher on Earth (75.3 h). At 60 % load the three groups stand in the same order: owners alone 1,144 to 1,630 h, the Earth-losing pooled regimes 80 to 484 h and the on-Moon regimes with the marketplace 45 to 77 h.

Table 5. Each decision regime at each load: mean urgent wait (h, censored at the horizon), its 95th percentile, the share of urgent items delivered within a day, tonnes delivered and left in the backlog at the end of the year, and whole-fleet utilisation. Cell means over the runs listed.

Load (%)	Regime	Runs	Urgent wait (h)	p95 (h)	Within 24 h	Delivered (t)	Backlog (t)	Fleet busy
10	A_EARTH plan_ahead	80	1,159.8	5,296	50 %	293	160	1.9 %
	A_EARTH strict	80	1,149.9	5,239	28 %	291	162	1.9 %
	A_MOON	80	1,144.2	5,230	51 %	293	160	1.8 %
	B_EARTH strict	80	75.3	201	49 %	449	4	2.3 %
	B_EARTH plan_ahead	80	18.8	84	83 %	453	1	2.7 %
	B_EARTH_RELAY 0.8	80	13.0	53	83 %	454	1	2.5 %
	C_AUCTION†	30	11.9	47	84 %	449	0	2.5 %
	B_EARTH_RELAY 1.0†	30	11.9	43	84 %	449	0	2.5 %
	O_ORACLE†	30	11.9	43	84 %	449	0	2.5 %
	C_LEADER†	30	11.5	42	84 %	450	0	2.5 %
	C_LEARNED_LIN†	30	11.0	40	86 %	450	0	2.5 %
	C_LEARNED_MLP†	30	10.9	41	86 %	448	0	2.5 %
30	A_EARTH strict	80	1,256.6	6,257	24 %	943	524	6.0 %
	A_EARTH full coverage†	30	1,247.4	6,292	50 %	956	513	6.1 %
	A_EARTH plan_ahead	80	1,233.1	6,451	48 %	955	513	6.1 %
	A_MOON	80	1,217.1	6,364	49 %	955	513	6.0 %
	B_EARTH strict	80	104.9	289	39 %	1,454	18	7.0 %
	B_EARTH plan_ahead	80	22.3	110	80 %	1,470	2	8.2 %
	B_EARTH_RELAY 0.8	80	17.6	69	78 %	1,470	2	7.5 %
	B_EARTH_RELAY 1.0†	30	14.9	66	82 %	1,469	1	7.6 %

	O_ORACLE†	30	14.9	66	82 %	1,469	1	7.6 %
	C_LEADER†	30	14.7	64	83 %	1,471	2	7.6 %
	C_AUCTION hopper_refuel†	30	14.5	65	82 %	1,471	2	7.6 %
	O_ORACLE hopper_refuel†	30	14.3	60	82 %	1,469	1	7.6 %
	C_AUCTION†	30	14.1	62	83 %	1,470	2	7.6 %
	C_LEARNED_MLP†	30	13.8	64	83 %	1,471	2	7.6 %
	C_LEARNED_LIN†	30	11.8	52	86 %	1,471	2	7.5 %
60	A_EARTH strict	80	1,630.0	6,624	10 %	1,655	1,600	10.6 %
	A_EARTH plan_ahead	80	1,151.5	6,717	38 %	2,112	1,146	13.3 %
	A_MOON	80	1,145.4	6,554	36 %	2,110	1,148	13.0 %
	A_EARTH full coverage†	30	1,143.6	6,669	40 %	2,117	1,145	13.4 %
	B_EARTH strict	80	483.9	1,500	22 %	2,180	1,083	10.3 %
	B_EARTH plan_ahead	80	169.0	910	46 %	3,210	55	16.8 %
	B_EARTH_RELAY 0.8	80	79.8	249	45 %	3,230	36	15.3 %
	C_AUCTION†	30	76.7	279	55 %	3,253	19	15.7 %
	C_LEARNED_MLP†	30	75.3	306	57 %	3,250	23	15.7 %
	C_LEADER†	30	67.8	239	55 %	3,252	20	15.7 %
	B_EARTH_RELAY 1.0†	30	66.3	231	56 %	3,248	22	15.6 %
	O_ORACLE†	30	66.3	231	56 %	3,248	22	15.6 %
	C_LEARNED_LIN†	30	45.2	212	73 %	3,259	15	15.4 %

† Earth-independent regime: a seed gives the same run in every start year, so only the 30 runs of block one are counted and shown (change 4 of the pre-registration [43]).

5.2 Sharing: capacity and dispatch (H1)

Pooling cut the mean urgent wait by 1,232 h at the middle load (95 % CI 1,076 to 1,401 h, paired, n = 30), comparing the pooled dispatcher and owners alone, both with Earth always in view. H1 is supported (Holm-adjusted $p < 0.001$). At 60 % load the same comparison cut the wait by 1,077 h (95 % CI 984 to 1,175 h, n = 30, $p < 0.001$), so H1's second half, that pooling helps at 60 % load too, is supported (Table 7; the 60 % rows lie outside the Holm family of Section 4.5).

That difference lies almost entirely in cargo that owners alone never deliver. Owner C's small rovers and hoppers cannot lift its crates or its regolith lots, and without cross-owner pairs no owner can lift most team items; by the end of a year owners alone leave 513 t waiting. Counting delivered items only, which conditions on delivery and so compares different sets of items, owners alone with strict fallback wait 100 h longer than the full-information marketplace, and 99.5 % of the censored gap lies in items that owners alone had not delivered by the end of the year. Split as designed in Section 4.2, single-robot jobs pooled across owners, together with the pooled dispatcher, account for 844 h of the gap (68 %; 95 % CI 732 to 955 h) and lifts by pairs of different owners' rovers for 389 h (32 %; 301 to 483 h). The pooled side of the first part is the robots' auction, whose difference from the full-information marketplace is given in Section 5.4. By owner, pooling cut the mean urgent wait by 404 h for owner A, 631 h for owner B and 2,575 h for owner C, the owner without a cargo rover.

5.3 A dispatcher on Earth (H2)

A pooled dispatcher on Earth that stops when Earth sets kept urgent cargo waiting 89.9 h longer than the full-information marketplace, pooled over all start years (95 % CI 81.8 to 98.7 h, n = 80): 104.9 h against 15.0 h. The pre-registered margin was 10 % of the marketplace's mean, 1.5 h; the lower end of the interval clears it by a wide

distance in each of the six start years whose base visibility is below 60 % (2029, 2031, 2032, 2038, 2041 and 2044), where the year means of the difference range from 74 h to 113 h (Figure 1). H2 is supported (Holm-adjusted $p < 0.001$).

The backlog at the end of a blackout averages 47.8 t under strict fallback against 2.3 t for the marketplace, and the share of urgent items picked up within a day falls from 81 % to 39 %. Planning ahead recovers most of it: uploading per-vehicle queues and a standing rule for blackout landings leaves 7.3 h of extra wait (95 % CI 5.2 to 9.4 h). A relay that is up 80 % of the time leaves 2.6 h (1.7 to 3.4 h). That is outside the 10 % margin we set for it, so the secondary expectation that an 80 % relay comes within 10 % of the marketplace fails.

The original yardstick of H2, half the gap between owners alone and the marketplace, would have been 621 h, larger than anything Earth's availability could cost a pooled fleet; we report it as registered but do not test it (Section 4.6).

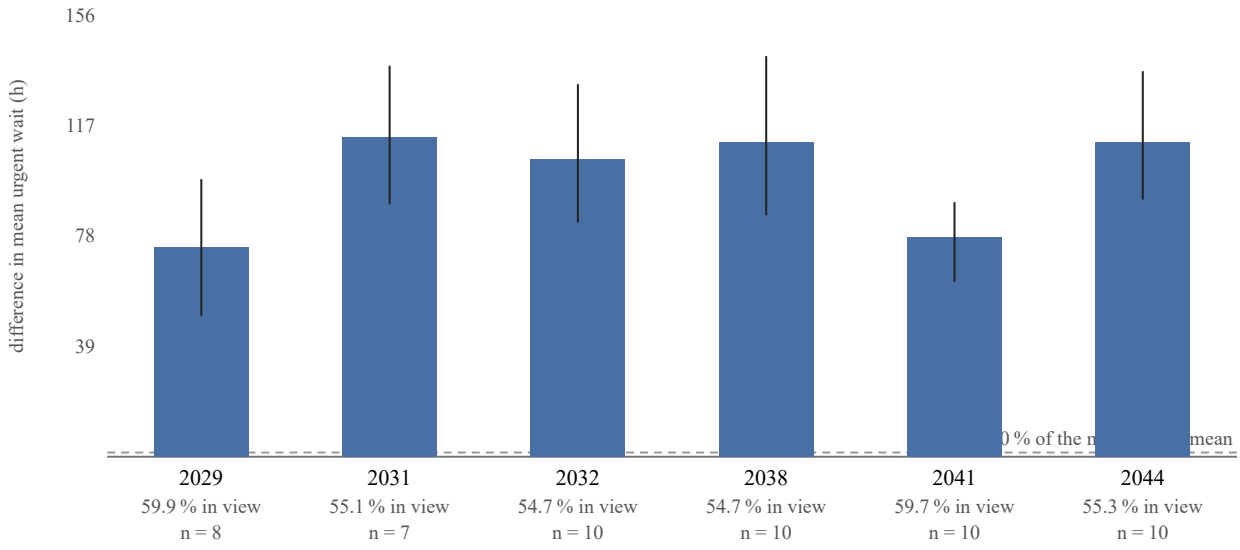


Figure 1. The strict dispatcher on Earth minus the full-information marketplace, mean urgent wait per start year with base visibility below 60 %, with 95 % bootstrap intervals and the pre-registered margin.

5.4 Deciding on the Moon (H3a)

The robots' own auction, at message loss 0.10 over the terrain-limited network, kept urgent cargo waiting 0.84 h less than the full-information marketplace (−5.6 %; 90 % CI −2.6 to 0.7 h, n = 30). We had registered the expectation that it would come within ±10 % of the marketplace, ±1.5 h. The 90 % interval is not inside that margin, so equivalence is not shown and H3a is not supported as stated; the interval includes zero, so the auction is shown neither better nor worse than the marketplace. Because the auction, the leader and the marketplace never wait for Earth, a seed gives the same run in every start year; this comparison therefore uses the 30 runs of block one only (change 4 of the pre-registration [43]). An earlier interim reading that counted the repeated runs as independent suggested the auction beat the marketplace; on independent runs that difference is not distinguishable from zero.

The elected leader, which runs the marketplace itself on the base computer and loses messages on the way, came within 0.26 h of it (−1.8 to 1.2 h).

5.5 Message loss, bundle depth and radio range (H3b, H3c)

With no message loss the four on-Moon regimes kept urgent cargo waiting 14.6 h (the elected leader), 14.3 h (the auction), 13.8 h (the learned network bids) and 10.8 h (the learned linear bids), against 14.9 h for the full-information marketplace on the same seeds; the learned linear bids lay below it by 4.2 h (2.7 to 5.7 h) and the others did not differ from it. Raising the loss to 0.40 added 1.0 h (−0.4 to 2.7 h) for the elected leader, 3.9 h (2.0 to 5.9 h) for the auction, 3.7 h (2.3 to 5.3 h) for the learned network bids and 4.3 h (2.4 to 6.4 h) for the learned linear bids, that is 0.18, 0.67, 0.67 and 1.00 % of the loss-free wait per point of loss, against about 0.3 % of makespan per point for the best allocators in Lott and Honary's benchmark [11]. The auction and the learned network bids rose above the marketplace at loss 0.40 (one-sided 95 % lower bound of the paired difference above zero); the elected leader and the learned linear bids stayed level with or below it through loss 0.40. Bursty loss at a mean rate of 0.25, against steady loss at the same rate, cut the wait of the auction by 1.4 h (95 % CI 0.4 to 2.4 h) and cut the wait of the learned linear bids by 1.2 h (95 % CI 0.4 to 2.2 h) and left the others within their intervals of zero.

Bundles of up to five jobs hurt the auction: against depth 1, depth 5 raised the urgent wait by 6.0 h (95 % CI 4.4 to 7.8 h) at loss 0.25 and raised it by 10.0 h (95 % CI 7.7 to 12.6 h) at loss 0.40, and depth 2 raised it by 3.7 h (95 % CI 1.8 to 5.6 h) at loss 0.25 and raised it by 5.0 h (95 % CI 2.3 to 7.6 h) at loss 0.40. Stale claims, bids on items that another robot had already taken by the time the bundle was accepted, rose from 0 to 314 per year at depth 5 and loss 0.40. H3b, that depth 5 is worse than depth 1 at loss 0.25 to 0.40, is supported. For the learned bids at loss 0.40, depth 5 raised the wait of the learned network bids by 9.8 h (95 % CI 7.6 to 12.0 h) and raised the wait of the learned linear bids by 8.7 h (95 % CI 6.4 to 11.1 h).

Radio range mattered less than loss: against the default 5 km range, a 2 km range lowered the auction's urgent wait by 0.1 h (95 % CI −0.2 to 0.0 h), not distinguishable from zero, a 10 km range left it unchanged on every seed and a radio that reaches every robot raised it by 1.2 h (95 % CI −0.2 to 2.7 h), not distinguishable from zero. With the lost-link rule off the wait rose by 3,267.5 h (95 % CI 3,011.7 to 3,540.4 h), and the share of robot-time out of reach of every mast went from 5.1 % to 83.1 % (809 returns to a mast per year with the rule on). A second mast at pad B raised the wait by 1.5 h (95 % CI 0.1 to 3.2 h) and cut that share from 5.1 % to 5.0 %. H3c, that the elected leader matches the auction unless the mesh partitions, was tested in ten cells by equivalence within ±10 % (90 % CI of the ratio): the leader matched the auction under loss 0 and a perfect radio; it differed under loss 0.40 (faster by 2.6 h, 95 % CI 0.7 to 4.5 h) and the lost-link rule off (faster by 871.6 h, 95 % CI 571.3 to 1,149.8 h); under loss 0.10, loss 0.25, bursty loss, a 2 km range, a 10 km range and the second mast the interval was too wide to decide either way. E41's default auction cell reran E40's auction at 30 % on the same 30 seeds and reproduced every run exactly.

5.6 Learned bids (H4)

The learned linear correction cut the mean urgent wait relative to the hand-built auction by 2.3 h at the middle load (16.5 %; 95 % CI 0.6 to 3.9 h, n = 30) and by 0.9 h at light load (7.9 %; −0.1 to 2.2 h). For the linear model H4's first part is supported (Holm-adjusted p = 0.017); its second part, no difference beyond ±2 % at light load, fails. At 60 % load the linear model cut the wait by 31.5 h (95 % CI 25.2 to 38.2 h) and the network's difference, −1.3 h (95 % CI −6.0 to 3.1 h), includes zero (n = 30; Table 7).

The network did not improve on the hand-built bids. At the middle load its difference from the auction was −0.25 h (95 % CI −1.6 to 1.1 h), and at light load −1.06 h (−3.0 to 0.4 h); neither interval excludes zero. H4 as registered names the learned policy without saying which model; for the network it is not supported.

The largest weights lower the bid, and so favour the item, for the planned hours of the job (+1.93 h per unit, the feature scaled by 1/3), the load as a share of capacity (+1.28 h per unit) and the number of robots heard (+1.04 h per unit), and raise it for the slack to the item's target time (−1.96 h per unit, the feature scaled by 1/10), the hours to drive to the pad (−1.41 h per unit) and the battery's state of charge (−1.39 h per unit); the other weights in each direction are smaller than these. The gain does not come from other cargo: in the guard runs at 60 % load, delays of normal-priority and bulk cargo and tonne-days waiting all fell with the learned bids (−15 % for tonne-days), and tonnes delivered did not change. Outside its training range, at two and five owners, message loss 0.25 and 0.40, light load and start year 2038, the linear model kept the same sign in all six cells, with intervals clear of zero in three of them (Table 6). A fleet twice as large could not be run in our time budget.

Table 6. The linear bids outside their training range, from the evaluation of the learned bids before the main runs: mean urgent wait (h) with hand-built and learned bids, and the paired difference with its 95 % interval, on ten evaluation seeds the model never saw. 2eq and 5eq split the same eight vehicles and the demand equally among two and five owners. Where E41 reran a cell on 30 seeds (loss 0.25 and 0.40, Section 5.5) and E40 on 30 seeds (load 10 %, Section 5.6), those values supersede these.

Cell	Hand-built (h)	Learned (h)	Difference (h)	n
2eq	16.0	11.4	−4.59 [−8.24, −0.95]	10
5eq	14.0	12.6	−1.33 [−3.50, 0.84]	10
loss 0.25	16.4	13.4	−3.07 [−4.42, −1.72]	10
loss 0.40	18.0	16.4	−1.62 [−4.91, 1.68]	10
load 10 %	11.5	10.3	−1.22 [−2.12, −0.31]	10
year 2038	13.3	12.0	−1.23 [−3.91, 1.45]	10

Table 7. The confirmatory family at the middle load: paired mean difference with its 95 % bootstrap interval, the two-sided bootstrap p and Holm's adjusted p. H3a is an equivalence claim and is tested in Section 5.4. The last three rows are the registered 60 % halves of H1 and H4, outside the Holm family, with the raw p only.

Hypothesis	Comparison	Block	n	Difference [95 % CI]	p	Holm p	Outcome
H1	H1: B_EARTH_RELAY 1.0 - A_EARTH full coverage	v1	30	−1,232.4 h [−1,401.0, −1,075.5]	< 0.001	< 0.001	supported
H2	H2: B_EARTH strict - O_ORACLE, pooled years	all	80	+89.9 h [81.8, 98.7]	< 0.001	< 0.001	supported
H4, linear	H4: C_LEARNED_LIN - C_AUCTION	v1	30	−2.3 h [−3.9, −0.6]	0.009	0.017	supported
H4, network	H4: C_LEARNED_MLP - C_AUCTION	v1	30	−0.2 h [−1.6, 1.1]	0.73	0.73	not supported
H6	H6: cross-owner - same owner, team 25, loss 0.1	v1	30	+5.4 pp [3.5, 8.5]	< 0.001	< 0.001	supported
H1, 60 %	H1: B_EARTH_RELAY 1.0 - A_EARTH full coverage	v1	30	−1,077.3 h [−1,174.9, −984.4]	< 0.001	–	supported
H4, linear, 60 %	H4: C_LEARNED_LIN - C_AUCTION	v1	30	−31.5 h [−38.2, −25.2]	< 0.001	–	supported
H4, network, 60 %	H4: C_LEARNED_MLP - C_AUCTION	v1	30	−1.3 h [−6.0, 3.1]	0.56	–	not supported

5.7 Team lifts across owners (H6)

Allowing pairs of different owners raised the share of team items delivered from 12 % to 93 % at team share 0.10 (paired difference 82 percentage points, 95 % CI 78 to 86, n = 30) and from 11 % to 96 % at team share 0.25 (paired difference 86 percentage points, 95 % CI 83 to 88, n = 30) at message loss 0.10, and the share of all items delivered by 1.7 percentage points (1.5 to 2.0) at team share 0.10 and 5.4 percentage points (3.5 to 8.5) at team share 0.25. H6's first part, registered on the share of all items at team share 0.25, is supported (Holm-adjusted p < 0.001, Table 7). The gain held at loss 0.25 and 0.40 (Table 8).

Failed meetings rose with message loss: at team share 0.25 the auction with the commit step and a 1 h timeout had on average 4.0 failed meetings per year at loss 0.10, 12.1 at 0.25 and 25.2 at 0.40. Without the commit step it

had more at loss 0.25 (15.2) and more at 0.40 (35.4): paired by seed, the step cut failed meetings by 3.1 (0.6 to 5.8) at loss 0.25 and 10.2 (4.5 to 16.0) at loss 0.40, so the second part of H6, that failed meetings rise without a commit step, is supported. The step is not free: the urgent wait with it was 1.8 h (0.1 to 3.6 h) longer at loss 0.25 and 9.1 h (6.8 to 11.5 h) longer at loss 0.40, a cost that excludes zero at both losses, because a meeting that the step refuses is a lift that waits. The timeout sets the waiting without buying fewer failures: at loss 0.25 the hours robots spent waiting at meeting points were 6.5 with a 0.5 h timeout, 12.1 with 1 h and 27.7 with 2 h, while failed meetings stayed between 12.1 and 13.8. The exploratory variant in which the base computer arbitrates each meeting had no failed meetings at either loss.

Table 8. Every E42 cell at the middle load: mean urgent wait (h), tonnes delivered, team items delivered out of those offered, failed meetings, meetings with no agreement and the hours robots waited at meeting points, all per simulated year.

Cell	n	Urgent wait (h)	Delivered (t)	Team items	Failed meetings	No agreement	Meeting wait (h)
A_EARTH strict own teams team 00	30	930.3	944	0.1 of 1.2	0.0	0.0	0.0
A_EARTH strict own teams team 10	30	1,262.6	943	0.5 of 12.4	0.0	0.0	0.0
A_EARTH strict own teams team 25	30	1,772.3	940	0.8 of 28.4	0.0	0.0	0.0
C_AUCTION team 00 any loss 0	30	10.5	1,447	1.2 of 1.2	0.0	0.0	0.0
C_AUCTION team 00 any loss 0.1	30	10.2	1,447	1.2 of 1.2	0.4	1.2	0.4
C_AUCTION team 00 any loss 0.25	30	10.8	1,447	1.2 of 1.2	1.4	5.9	1.4
C_AUCTION team 00 any loss 0.4	30	10.8	1,447	1.2 of 1.2	2.3	15.7	2.3
C_AUCTION team 00 same_owner loss 0	30	23.3	1,444	0.2 of 1.2	0.0	0.0	0.0
C_AUCTION team 00 same_owner loss 0.1	30	23.2	1,444	0.2 of 1.2	0.1	0.2	0.1
C_AUCTION team 00 same_owner loss 0.25	30	23.3	1,444	0.2 of 1.2	0.4	1.5	0.4
C_AUCTION team 00 same_owner loss 0.4	30	23.8	1,443	0.2 of 1.2	0.3	2.5	0.3
C_AUCTION team 10 any loss 0	30	14.3	1,471	11.7 of 12.4	0.0	0.0	0.0
C_AUCTION team 10 any loss 0.1	30	14.1	1,470	11.6 of 12.4	2.2	5.4	2.2
C_AUCTION team 10 any loss 0.25	30	15.7	1,471	11.6 of 12.4	7.3	27.7	7.3
C_AUCTION team 10 any loss 0.4	30	18.2	1,471	11.7 of 12.4	11.1	88.8	11.1
C_AUCTION team 10 same_owner loss 0	30	405.6	1,443	1.5 of 12.4	0.0	0.0	0.0
C_AUCTION team 10 same_owner loss 0.1	30	402.9	1,443	1.5 of 12.4	0.4	0.7	0.4
C_AUCTION team 10 same_owner loss 0.25	30	399.1	1,443	1.5 of 12.4	1.0	3.6	1.0
C_AUCTION team 10 same_owner loss 0.4	30	402.7	1,443	1.5 of 12.4	1.4	11.8	1.4
C_AUCTION team 25 any loss 0	30	19.8	1,505	27.0 of 28.4	0.0	0.0	0.0
C_AUCTION team 25 any loss 0.1	30	21.5	1,506	27.4 of 28.4	4.0	9.8	4.0
C_AUCTION team 25 any loss 0.1 arbitrated (exploratory)	30	19.8	1,505	27.0 of 28.4	0.0	0.0	0.0
C_AUCTION team 25 any loss 0.1 no commit	30	20.9	1,505	27.0 of 28.4	4.1	0.0	4.1
C_AUCTION team 25 any loss 0.1 timeout 0.5	30	21.3	1,504	26.5 of 28.4	4.2	10.7	2.1
C_AUCTION team 25 any loss 0.1 timeout 2	30	20.9	1,505	27.0 of 28.4	4.3	9.9	8.7
C_AUCTION team 25 any loss 0.25	30	23.2	1,503	26.4 of 28.4	12.1	50.8	12.1
C_AUCTION team 25 any loss 0.25 arbitrated (exploratory)	30	19.8	1,505	27.0 of 28.4	0.0	0.0	0.0

C_AUCTION team 25 any loss 0.25 no commit	30	21.3	1,503	26.6 of 28.4	15.2	0.0	15.2
C_AUCTION team 25 any loss 0.25 timeout 0.5	30	24.4	1,504	26.7 of 28.4	12.9	55.8	6.5
C_AUCTION team 25 any loss 0.25 timeout 2	30	25.5	1,505	26.9 of 28.4	13.8	58.7	27.7
C_AUCTION team 25 any loss 0.4	30	32.9	1,503	26.5 of 28.4	25.2	203.7	25.2
C_AUCTION team 25 any loss 0.4 arbitrated (exploratory)	30	19.9	1,505	26.9 of 28.4	0.0	0.0	0.0
C_AUCTION team 25 any loss 0.4 no commit	30	23.8	1,505	27.1 of 28.4	35.4	0.0	35.4
C_AUCTION team 25 any loss 0.4 timeout 0.5	30	30.6	1,506	27.3 of 28.4	26.0	203.8	13.0
C_AUCTION team 25 any loss 0.4 timeout 2	30	32.2	1,502	26.3 of 28.4	26.3	202.7	52.5
C_AUCTION team 25 same_owner loss 0	30	1,033.1	1,412	2.9 of 28.4	0.0	0.0	0.0
C_AUCTION team 25 same_owner loss 0.1	30	1,037.1	1,416	3.0 of 28.4	0.4	1.3	0.4
C_AUCTION team 25 same_owner loss 0.25	30	1,035.7	1,411	2.9 of 28.4	1.3	6.2	1.3
C_AUCTION team 25 same_owner loss 0.4	30	1,029.3	1,413	3.0 of 28.4	4.3	32.2	4.3

5.8 Sensitivities

Letting the hoppers refuel every month changed nothing that matters: the full-information marketplace's urgent wait moved by −0.61 h (−2.1 to 0.4 h).

5.9 What did not work

Imitating the full-information marketplace taught the bidding models nothing, because the marketplace uses the same bid rule: the hand-built bids already chose the marketplace's item in 92.9 % of held-out rounds, and every disagreement we examined (318 of 318 at 30 % load and 404 of 404 at 60 %) was an item the marketplace gave to another robot in the same round. The network's best tuning score came in its first generation; the linear model's came in its last. The network trained by search on its output layer did not beat the hand-built bids. The relay at 80 % coverage missed its equivalence margin. Bundle depth 5 made the auction worse at message loss 0.25 and 0.40 (Section 5.5). Of the 107 phase-5 checks of Section 3.11 tagged to this paper, 4 failed: D10 pairing A_EARTH plan_ahead (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): A_EARTH plan_ahead (E40, 10 % load, 3eq)), the condition did not hold; D10 pairing A_EARTH strict (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): A_EARTH strict (E40, 10 % load, 3eq)), the condition did not hold; D10 pairing A_MOON (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): A_MOON (E40, 10 % load, 3eq)), the condition did not hold; D10 pairing B_EARTH strict (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): B_EARTH strict (E40, 10 % load, 3eq)), the condition did not hold. The other 97 that ran passed; 6 were not run: A1 not applicable (not applicable: the code has no epsilon-auction; central assignment's optimality is D9); A3 not applicable (not applicable: the CBBA bound does not apply; the auction's own round limit and its agreement (no item carried twice) are A2 and A7); A4 not applicable (not applicable: the 50 % guarantee is CBBA's; not claimed); A5 not applicable (not applicable: DMG is the condition of CBBA's guarantees, which the papers do not claim; the learned correction adds to a bundle bid); L1 teacher agreement (no preregistered agreement threshold (PREREG H4 tests outcomes, not agreement); the numbers are reported in basis); L2 not applicable (not applicable: the learned policy is tuned to differ from its teacher; H4 tests the outcome); 1 met its

tolerance with a 99 % interval that misses the reference, recorded as a bias flag: D4 class 2. After every experiment had run, a verification check found that cargo ids were not unique: 9–18 of about 2,970 items a year reused an earlier id, in every regime (change 10 of the pre-registration). Ids enter the simulator's decisions only through a planning memo within one dispatch call. The fix was tested by rerunning 48 full years (E40, E43: every regime or mode, seeds 1–3), of which 48 gave outcomes identical item by item, compared by content without ids, 39 of them in this paper's experiment; reused ids fell from 670 to 0. No experiment was rerun and no reported number changes; trace and scene files written before the fix still carry the duplicate ids.

6. Discussion

6.1 What the results say about the question

We asked whether the gain from sharing vehicles among owners survives when the decisions move from Earth onto the robots. It does, and the measurements split the gain into three parts that behave differently. The first is capability. Almost the whole gap between owners working alone and a pooled fleet, 99.5 % of it at the middle load, is cargo that owners alone had not delivered by the end of the year: owner C's small vehicles cannot lift its crates, and no single owner can pair its robots for most team items (Section 5.2). Lifts by pairs of different owners' rovers account for 32 % of the gain, and single-robot jobs taken by another owner's rover, together with the pooled dispatcher, for the rest. The design cannot separate the dispatcher's share of that rest, but the pooled regimes that differ only in their dispatcher sit within a few hours of one another (Section 5.1), so it can be little. This part of the gain needs no dispatcher in any particular place. It needs an agreement that lets one owner's rover carry another owner's cargo and lets two owners' rovers lift together. The second part is where the pooled decision is made. Keeping it on Earth cost 89.9 h of mean urgent wait per year when the dispatcher stopped at every blackout, 7.0 times the wait under uninterrupted dispatch, and the cost appeared in every start year in which the base saw Earth less than 60 % of the time (Section 5.3). The Earth-bound dispatcher's loss comes from the blackouts. While Earth is down, nothing is assigned; vehicles finish their current job and wait, and cargo that lands in the meantime queues at the pads (Section 5.3). Uploading plans before each blackout left 7.3 h of it and an 80 % relay left 2.6 h, less, but more than our equivalence margin allows. Moving the decision onto the robots left none that the data could detect, though not none that they could exclude: the auction among the robots, which hear one another only over the terrain and lose one message in ten, kept urgent cargo waiting 0.84 h less than the full-information marketplace, a difference not distinguishable from zero and a small fraction of what the Earth-bound dispatcher lost (Section 5.4). The third part is the quality of the bids. A learned linear correction with 13 weights cut the auction's urgent wait by 16.5 % at the middle load, and a network trained the same way added nothing (Section 5.6). Beyond the three parts, Pairs of different owners' robots lifting heavy crates raised the share of team items delivered from 12 % to 93 %; a commit step before each meeting cut failed meetings under loss, at a cost in urgent wait (Section 5.7).

Not every registered expectation held, and the failures say something. The equivalence test of the auction against the marketplace is reported as it came out in Section 5.4: the point estimate is 0.84 h less than the marketplace, the 90 % interval −2.6 to 0.7 h, the margin ±1.5 h. Ten percent of a mean of 14.9 h is a narrow margin in hours, which is why an interval that includes zero can still fail to show equivalence, and why the practical reading is the one in the previous paragraph rather than a verdict on the hypothesis. The 80 % relay missed its margin for the same reason: its loss is small next to the strict dispatcher's and still larger than a tenth of the marketplace's mean. The network's failure to beat the hand-built bids says more about how we trained it than about networks (Section 7). Its training gave it less chance than the linear model: imitation left its hidden layers at their initial values and

the search adjusted only its output layer (Section 3.10), so this is no evidence that a network helps as trained here, and no evidence that a network cannot help.

6.2 Comparison with earlier studies

Studies of allocation under message loss compare an algorithm with itself under perfect links. The best decentralised allocators in Lott and Honary's benchmark lose 11–13 % of makespan between perfect delivery and 40 % loss [10, 11]. Our comparison is different in kind: the auction at 10 % loss is set against a dispatcher with perfect information, and the quantity is cargo delay, not makespan. On that harder comparison it differed from the reference by −5.6 %. The slope against loss, which is the quantity those benchmarks report, is measured in Section 5.5. NASA's CADRE rovers coordinate through an elected leader [31, 32]. Our leader regime, which runs the marketplace on the base computer over the same lossy radio, came within 0.26 h of the marketplace, so on this terrain, with a mast that most robots can see, the two on-Moon designs did not differ measurably at this loss, though equivalence could not be shown (Section 5.5). The marketplace is the auction's own bid rule given perfect information, which is why we call it a reference and not a bound (Section 5.4).

Transport research measures what fragmentation among operators costs when every vehicle can serve every customer: the cost is slack, and the smallest operator pays most of it [27, 28]. On the Moon the fleets are heterogeneous, and fragmentation removes capability, not only slack: 99.5 % of the gap we measured is cargo that owners alone never delivered. Kondor and colleagues found that the smallest operator pays most for fragmentation [28]; here the owner that pays most is the one without a cargo rover. Owner C's urgent cargo waited 2,575 h longer per year alone than pooled, against 404 h and 631 h for owners A and B (Section 5.2). What an owner lacks on the Moon is a vehicle class, not fleet size. Open-RMF lets robots of unrelated vendors share tasks and a traffic schedule, over a building network that is always up [30]; our regimes with an uninterrupted link to Earth are the lunar version of that assumption, and the strict regime is what happens when it fails. Most of what separates the pooled regimes is when cargo moves, while what separates them from owners alone is whether it moves at all (Section 5.1). Refuelling the hoppers changed nothing because a hopper carries 10 kg and the cargo that waits is heavy (Section 5.8).

Published learned allocators gain a few percent over their baselines: 1.3–5.3 % in deployed ride-hailing dispatch [23], and the learned bid inside CBBA of [18] is evaluated against CBBA's own bid under perfect links. Our 16.5 % is larger, and the reason is the baseline, not the learner. The linear correction beat not only the hand-built bids but the full-information marketplace that uses the same bid rule, so it measures the weakness of that rule under this demand; the hand-built rule was written, not tuned by search on the objective, while the linear model was. A stronger hand-built rule would shrink the gain, which is the comparison that Accorsi and colleagues ask for [26], and which we only partly made by testing the linear model against the network. What we add to that literature is the negative result for the network against a linear model with the same inputs, and the finding that imitation of the reference dispatcher teaches a bidder nothing when the reference uses the same rule. The linear weights say what the hand-built bid misses (Section 5.6). We could have seen that imitation would teach nothing from the definitions, before collecting 624,640 robot-item rows, and report it so others do not repeat it (Section 5.9).

6.3 Implications

For whoever runs a south-pole base, the results order the options. A relay to Earth buys less than moving the dispatch decision onto the robots: the relay at 80 % left 2.6 h of extra urgent wait, the robots' auction a difference not distinguishable from zero, and the robots' auction keeps working at the oxygen plant, which no relay reaches.

A dispatcher on Earth with plans uploaded before each blackout is an acceptable interim design at the lightest load, where every pooled regime except the strict one sits within 7.9 h of the others, and today's planned traffic sits at that load. The value of on-Moon dispatch grows with traffic.

For the owners, the agreement to share matters more than the software that shares. 99.5 % of the gain is cargo that owners alone never deliver, and it is available with any dispatcher that pools the fleet; what the dispatcher's place and quality add is the smaller sum measured in Sections 5.3 to 5.6. Who pays whom for a lift, and what keeps an owner from misreporting its costs, are contract and market questions, taken up in the companion paper on the vehicles, the network and the market [44]. For the designers of LunaNet [39], the traffic that matters here is robot-to-robot: bids, awards, meeting commitments and relays of all three, which no current specification standardises; the message volumes and the computer each robot needs are measured in [44]. The gain from learning needed no neural network: it came from 13 weights an operator can read, and the safety check on each vehicle decides whether a move is allowed whatever the bids say [45]; a lunar fleet does not need an opaque planner to get the gain we measured.

6.4 Limitations

Section 7 sets out the limitations in full. The main ones are that the radio model is geometric, that one site, one terrain tile and one demand model are simulated, that the robots' lost-link rule is our assumption, that the full-information marketplace is a reference and not an optimal dispatcher, and that the network was trained with less freedom than the linear model. The comparisons between regimes that never wait for Earth rest on 30 paired seeds, which is enough to show the differences we found and not enough to show equivalence within 10 % of a small mean.

6.5 Future work

Three extensions follow directly. The first is to train the network end to end on the simulated objective and from the linear policy's starting point, so that the question of whether nonlinearity buys anything is answered rather than left open. The second is a second site and a second terrain tile, with a link budget in the radio model, to learn whether the auction's result depends on the Connecting Ridge's mast coverage. The third is a hardware test: the bidding, the meeting protocol and the lost-link rule on rovers of the CADRE class over a real radio on an analogue site, which would measure what our simulation cannot, the message losses and timings of real links. Beyond these, the companion papers take the roads, the separation rules and the safety check [45], and the payments, the audits and the hardware sizing [44].

7. Limitations

The radio model is geometric. A link exists when two antennas see each other within a set range, with no link budget, no fading and no interference, and the range and clearance are provisional values that Section 5.5 varies. Real links on the ridge will fail in ways this model does not produce, and the message-loss rates we impose stand in for all of them.

One base, one terrain tile, one demand model. The base layout, the terrain, the cargo mix and the arrival process are those of our earlier papers [40–42]; three levels of demand are simulated, of which the lightest is the closest to today's plans. Nothing here says how the results change at another site, and the mast's coverage of the ridge is part of why the on-Moon regimes work.

The lost-link rule is ours. Robots that nobody hears drive to a rally point that the base's mast can see (Section 3.9). Without that rule the robots' network falls apart on this terrain, and the rule's cost is part of every on-Moon result. A base would need some such rule, and it need not be ours.

The marketplace is a reference, not a bound. It is the auction's own bid rule given perfect information and no dependence on Earth; it is not an optimal dispatcher, and the learned linear bids beat it. “Within 10 % of the marketplace” therefore means within 10 % of a good hand-built dispatcher, not of the best possible one.

Owners are honest and roads are free. Owners report their costs truthfully and accept the assignments they win, and vehicles do not interfere with one another on the road. Both assumptions are removed in the companion papers [44, 45], and the delay that rules of the road add is measured there.

The network was given a narrow test. Imitation left the network's two hidden layers at their random initial values, and the search on the objective adjusted only its 32 output weights, while the linear model's search covered all 13 of its weights (Section 3.10). The network was therefore a linear model on fixed random features of the inputs, searched with fewer candidates per weight than the linear model. Its failure to beat the hand-built bids is no evidence that a network helps as trained here; it is no evidence that a network cannot help. Two tests would settle it: training all 1,537 weights end to end on the simulated objective, with a gradient-free method such as an evolution strategy or with a policy-gradient method, and starting the network from the learned linear policy, so that it begins at the linear model's result and can only add what nonlinearity buys. Neither was run, and the pre-registration froze both models before the main runs (Section 4.6).

The simulation is checked, not validated against a lunar fleet. The engine's checks (Section 3.11) and those of our earlier papers compare it with closed-form cases, degenerate settings and its own earlier versions. No multi-owner fleet operates on the Moon, so no measured delay exists to validate against, and the absolute waits in Table 5 should be read as model outputs under stated assumptions; the paired differences between regimes are the result.

Censored waits. Cargo not delivered by the end of a simulated year carries the wait it had accumulated by then, so the primary outcome is a censored mean that understates the true wait of what owners alone leave behind, and the delivered-only mean reported in Section 5.2 conditions on delivery and so compares different sets of items. We report both and test neither as a substitute for the other; the treatment is the analogue, for a simulation, of reporting how missing observations were handled.

Three defects were found in the runs and corrected before the final analysis. Regimes that never wait for Earth give the same run for a seed whatever the start year, so their second block repeated seeds; comparisons between such regimes use the first block only (n = 30), and an interim reading that counted the repeats is withdrawn (Section 4.6). The auction's ordering of robots depended on a per-process hash, which made runs under message loss valid draws but not bit-reproducible; the order was fixed and the affected runs repeated, and the repeated runs changed no result. The third, in the engine's event handling, is described with its before-and-after comparison in Section 4.6. All three are recorded with dates in the pre-registration file [43].

The registered seed rule was not applied to H3a. Its 90 % half-width exceeded the 10 % threshold that called for ten more seeds, and we report the equivalence test as run rather than extend the sample after seeing the result (Section 4.5).

8. Conclusion

This paper asked whether the gain from pooling several owners' vehicles at a lunar south-pole base survives when dispatch moves from Earth onto the robots. On a simulated base with real Earth-link windows and terrain, it does. A pooled dispatcher that stops when Earth sets costs 89.9 h of mean urgent wait, a cost that appears in every year of poor Earth visibility; planning ahead and a relay recover most of that; an auction among the robots, at 10 % message loss, came within 0.8 h of the full-information marketplace, a difference not distinguishable from zero but too wide to show equivalence. Almost all of the value of pooling, 99.5 % of the gap, is cargo that owners alone never deliver; lifts by two owners' rovers account for 32 % of it. A learned correction to the bids with 13 weights cut the wait by a further 16.5 %; a small network trained the same way did not.

For practice, the order of value is an agreement to share, then dispatch by the robots on the Moon, then a relay to Earth, then a learned bid. For research, the negative results, a network no better than a linear model and an imitation that teaches nothing, count as much as the positive ones; the pre-registered plan shows which is which. The results hold for one base and terrain, a geometric radio and honest owners. A fleet of several owners can work together without Earth. What it needs most is permission.

Data and code availability

The simulation engine, the decision regimes, the learned policies, the experiment runner, every results file this paper quotes and the script that builds the paper from them are in the repository at github.com/azizjonrahimov/Lunar-Haul (directories `lunarsim/atc`, `experiments/phase5`, `results` and `paper/p1`), released under the MIT licence. The paper is rebuilt from the results files with `python paper/p1/build_p1.py`. The pre-registration with its dated changes is `paper/atc/PREREG.md` [43]. The runs are reproducible from the seeds and settings listed in Section 4; the policy files record their training settings. The live site at azizjonrahimov.github.io/Lunar-Haul shows the base, the vehicles and the results.

Author contributions

A.R. set the questions, the scope of the three papers and the standards of evidence, directed the design of the simulation and the experiments, reviewed the hypotheses, the results and the text, and is responsible for the content.

Use of AI tools

Large-language-model assistants (Anthropic's Claude, used through Claude Code) wrote the simulation code, the experiment runner and the analysis scripts, ran the experiments, drafted and revised the text and compiled the reference list, all under the author's direction and review. Every number in the paper is written by the build script from the results files, and each sentence that states a direction or a verdict is checked by that script against the data; no number was typed from an assistant's output. The references were collected from published sources and the repository's research notes; entries not yet checked against the publisher's record are marked in the notes and will be checked before submission. The author takes responsibility for the whole.

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No external funding was received. The author founded Roatify, which proposes a traffic and dispatch network for lunar vehicles of the kind simulated here. The pre-registration, the negative results and the build-script checks are the safeguards offered against that interest.

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